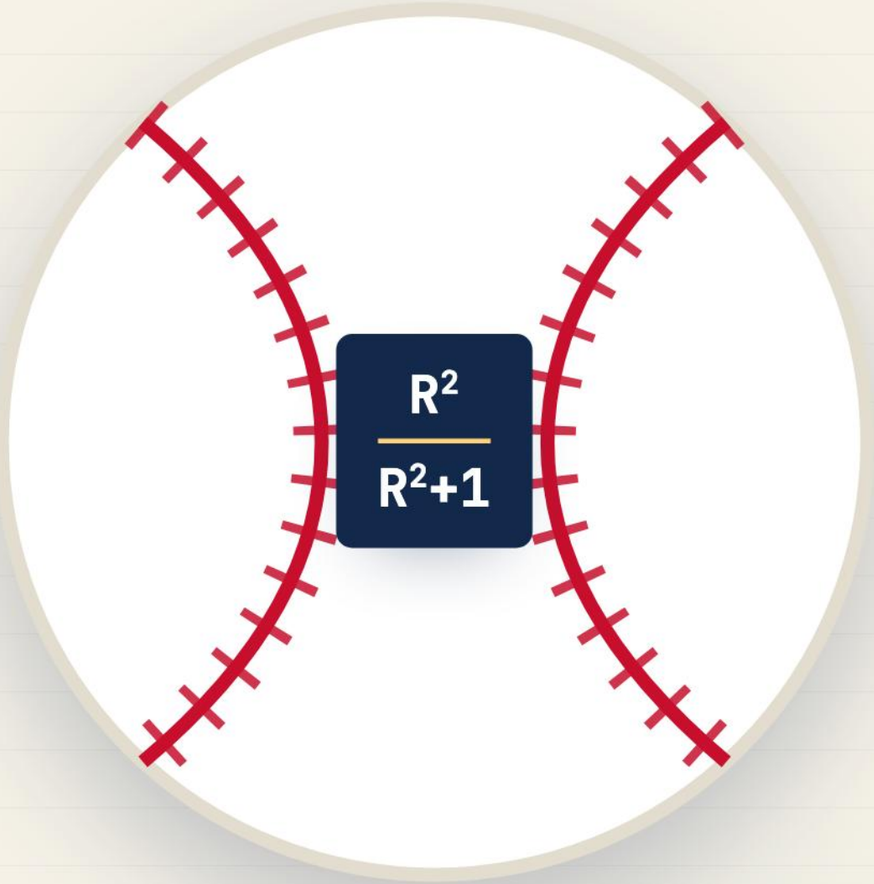
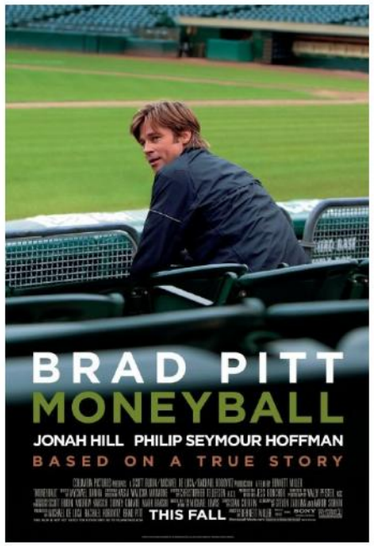
SPM 261 · WEEK 9-1

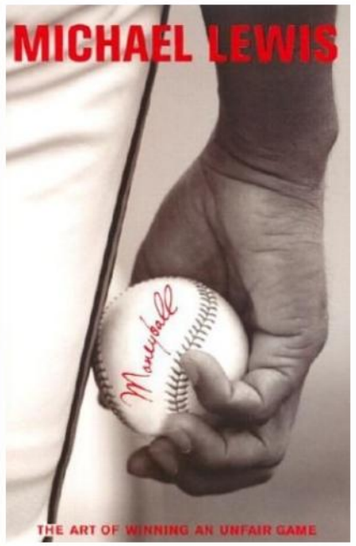
# BASEBALL ANALYTICS:

## MATHLETICS CH.1

SPM261 Intro to Sport Management











# SABERMETRICS

http://m.mlb.com/glossary/advanced-stats








https://www.youtube.com/watch?v=JXyisZLObHc&list=PLZbXA4lyCtqosSLgdOSwUWDK_KH9c2rIq




https://www.youtube.com/watch?v=aNDj-H1jxV0&list=PLZbXA4lyCtqosSLgdOSwUWDK_KH9c2rIq&index=2



# WHAT IS SPORT ANALYTICS?

01 Data

02 Analyzing skill

03 Data interpretation

04 Basic features of the data

05 Data visualization

06 Prediction-significance test

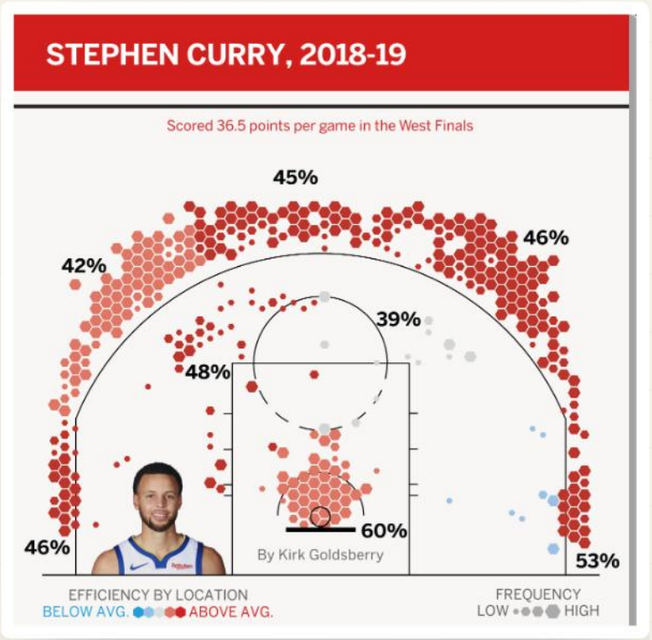
SPORT ANALYTICS

# WHAT IS SPORT ANALYTICS? CONT.

Player/team performance, winning/ performance prediction, fan emotions/reactions (i.e., social media), economic impact, betting market, rule/ policy test, playoff team selection, etc.



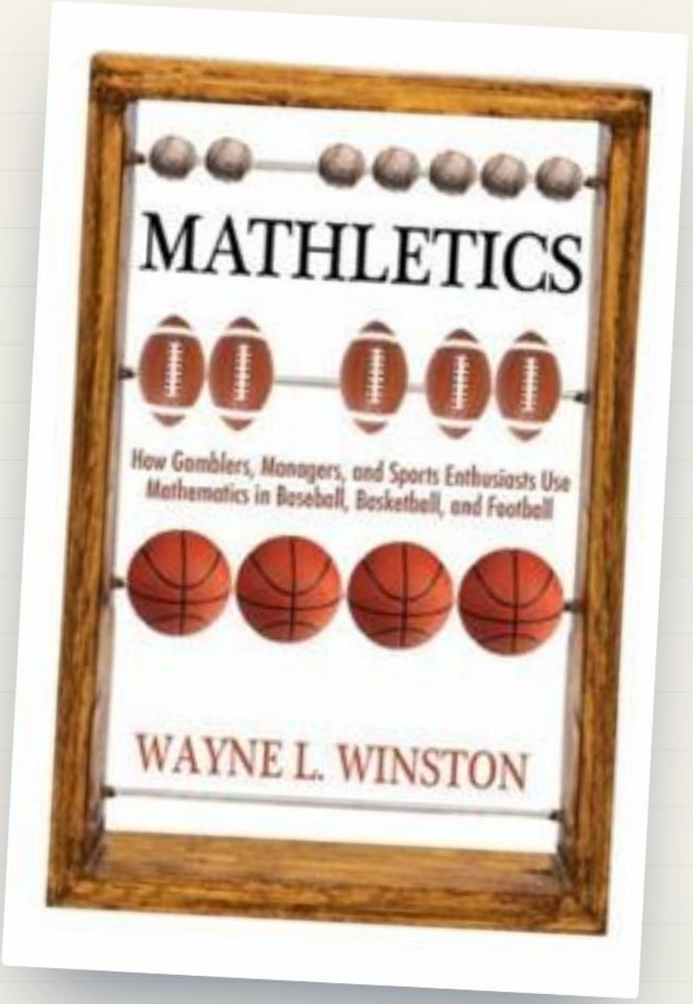

https://www.youtube.com/watch?v=MpLHMKTolVw



# PLEASE READ THE BOOK MATHLETICS:

**(a) Ch.1 Baseball's Pythagorean Theorem**

You can download the book Mathletics on course website. See Module 11






MATHLETICS · CHAPTER 1

# BASEBALL'S PYTHAGOREAN THEOREM (CH.1)




# PREDICTING A TEAM'S WINNING PERCENTAGE & # OF GAMES

PYTHAGOREAN THEOREM

# BASEBALL'S PYTHAGOREAN THEOREM (CH.1) CONT.

- The more runs a baseball team scores, the more games the team should win.
- Bill James introduces a formula that measures percentage of games won by baseball team (p.3).
- Pythagorean Theorem states that a triangle is a right triangle if and only if $a^2+b^2=c^2$.
- For example, a triangle with sides of lengths 3, 4, and 5 is right triangle because $3^2+4^2=5^2$.

$$\frac{\text{runs scored}^2}{\text{runs scored}^2 + \text{runs allowed}^2} = \text{estimate of percentage of games won.} \quad (1)$$

This formula has several desirable properties.

- The predicted win percentage is always between 0 and 1.
- An increase in runs scored increases predicted win percentage.
- A decrease in runs allowed increases predicted win percentage.

**If R (scoring ratio) =** $\frac{\text{runs scored}}{\text{runs allowed}}$

**The equation (1) can be interpreted as**

$$\frac{R^2}{R^2+1} = \text{estimate of percentage of games won}$$

# BASEBALL'S PYTHAGOREAN THEOREM (CH.1) CONT.

## WHY?

The equation (1) is same as the following fraction (if you divide the numerator and denominator of (runs allowed)$^2$:

$$\frac{\frac{\text{runs scored}^2}{\text{runs allowed}^2}}{\frac{\text{runs score}^2}{\text{runs allowed}^2} + \frac{\text{runs allowed}^2}{\text{runs allowed}^2}}$$

**If R (scoring ratio) =** $\frac{\text{runs scored}}{\text{runs allowed}}$

$$\frac{R^2}{R^2 + 1}$$ **= estimate of percentage of games won**

# BASEBALL'S PYTHAGOREAN THEOREM (CH.1) CONT.

**Figure 1.1 shows how well $\frac{R^2}{R^2+1}$ predicts MLB team's winning percentage.**

1. For example, the 2006 Detroit Tigers scored 822 runs and gave up 675 runs.
2. Their scoring ratio was R (scoring ratio) = $\frac{822}{675}$ = **1.218**
3. Their predicted win percentage was $\frac{1.218^2}{1.218^2+1}$ = **.597**
4. Actual win percentage (win/total games played) was **.586**
5. Absolute error (actual win % - predicted win %)
   So Tigers absolute error was |.586-.597| = **0.011**

MAD = 0.020

| Year | Team | Wins | Losses | Runs scored | Runs allowed | Scoring ratio | Predicted winning % | Actual Winning % | Absolute Error |
|---|---|---|---|---|---|---|---|---|---|
| 2006 | Diamondbacks | 76 | 86 | 773 | 788 | 0.981 | 0.490 | 0.469 | 0.021 |
| 2006 | Braves | 79 | 83 | 849 | 805 | 1.055 | 0.527 | 0.488 | 0.039 |
| 2006 | Orioles | 70 | 92 | 768 | 899 | 0.854 | 0.422 | 0.432 | 0.010 |
| 2006 | Red Sox | 86 | 76 | 820 | 825 | 0.994 | 0.497 | 0.531 | 0.034 |
| 2006 | White Sox | 90 | 72 | 868 | 794 | 1.093 | 0.544 | 0.556 | 0.011 |
| 2006 | Cubs | 66 | 96 | 716 | 834 | 0.859 | 0.424 | 0.407 | 0.017 |
| 2006 | Reds | 80 | 82 | 749 | 801 | 0.935 | 0.466 | 0.494 | 0.027 |
| 2006 | Indians | 78 | 84 | 870 | 782 | 1.113 | 0.553 | 0.481 | 0.072 |
| 2006 | Rockies | 76 | 86 | 813 | 812 | 1.001 | 0.501 | 0.469 | 0.031 |
| **2006** | **Tigers** | **95** | **67** | **822** | **675** | **1.218** | **0.597** | **0.586** | **0.011** |
| 2006 | Marlins | 78 | 84 | 758 | 772 | 0.982 | 0.491 | 0.481 | 0.009 |
| 2006 | Astros | 82 | 80 | 735 | 719 | 1.022 | 0.511 | 0.506 | 0.005 |
| 2006 | Royals | 62 | 100 | 757 | 971 | 0.780 | 0.378 | 0.383 | 0.005 |
| 2006 | Angels | 89 | 73 | 766 | 732 | 1.046 | 0.523 | 0.549 | 0.027 |
| 2006 | Dodgers | 88 | 74 | 820 | 751 | 1.092 | 0.544 | 0.543 | 0.001 |
| 2006 | Brewers | 75 | 87 | 730 | 833 | 0.876 | 0.434 | 0.463 | 0.029 |
| 2006 | Twins | 96 | 66 | 801 | 683 | 1.173 | 0.579 | 0.593 | 0.014 |
| 2006 | Yankees | 97 | 65 | 930 | 767 | 1.213 | 0.595 | 0.599 | 0.004 |

Figure 1.1. Baseball's Pythagorean Theorem, 1980–2006. See file Standings.xls.

EXERCISE

# EXERCISE (1-POINT)

**2019 MLB_Student Excel file on course website (see Module 11) or link below**

https://drive.google.com/file/d/1Qgk1WN_ky61AyTZa6ckXWMxrD6As2F78/view?usp=sharing

Complete the following columns: Scoring Ratio, Predicted Winning %, Actual Winning %, Absolute Error

| SCORING RATIO | PREDICTED WINNING % | ACTUAL WINNING % | ABSOLUTE ERROR |
|---|---|---|---|

- Upload your completed answer to Module 11 (see In-Class Assignments- Baseball Analytics)
- This will replace your handwritten note.

IN-CLASS ASSIGNMENT

# IN-CLASS ASSIGNMENTS (1-POINT)

## HOW DO WE USE THIS INFORMATION?

On July 4, 2005, the Nationals were in first place with a record of **50-32**

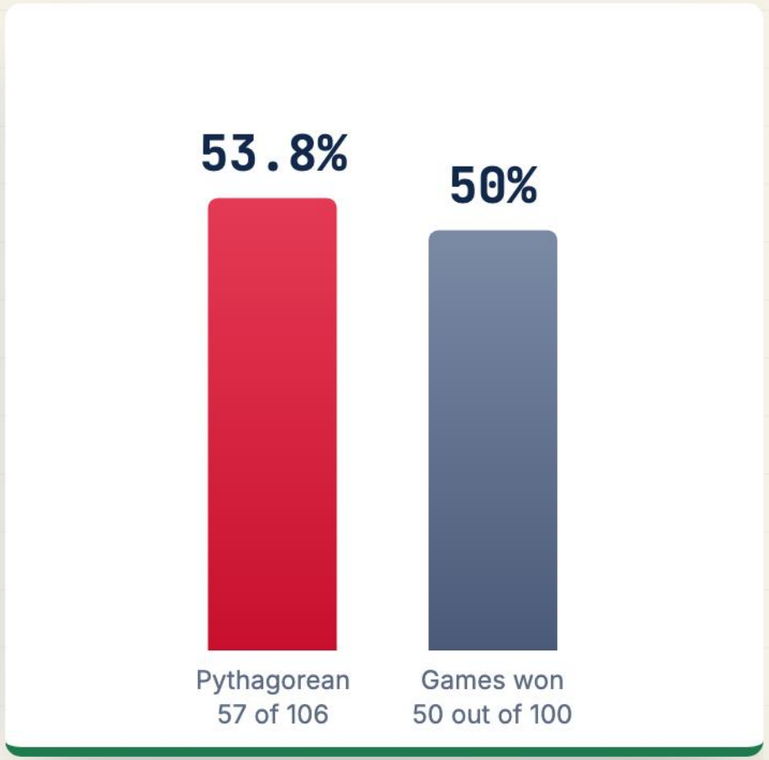
On July 4, 2005, the Nationals scoring ratio was **.991**

- Calculate the 2005 Nationals' predicted season records (# of wins and losses) using the scoring ratio. MLB has a total of 162 games per season.
- In 2005, National finished XX (# of win) – XX (# of loss)?
- This will replace your handwritten note.
- Please download the form (from Module 11) and return your answer with the formula.

# BASEBALL'S PYTHAGOREAN THEOREM (CH.1) CONT.

## HOW WELL DOES THE PYTHAGOREAN THEOREM FORECAST?

- Compared the Pythagorean Theorem's forecast for each MLB playoff series (1980-2007) against a prediction based just on games won.
- For each playoff series the Pythagorean method would predict the winner to be the team with the higher scoring ratio, while the games won approach simply predicts the winner of a playoff series to be the team that won more games.
- Pythagorean approach correctly predicted 57 of 106 playoff series (53.8%), while the games won approach correctly predicted the winner of only 50% (50 out of 100) playoff series.
- Regular season is a relatively poor predictor of the playoffs in baseball because a team's regular season record depends on the performance of five starting pitchers.
- During the playoffs teams only use three or four starting pitchers, so much of the regular season data are not relevant for predicting the outcome of the playoffs.





# QUESTIONS?

SPM 261 · BASEBALL ANALYTICS